

Computational Control Certificate

CTRL-03 — Exact integer maximization

Erdős #81, Paper II — internal validation evidence

Reference. Paper II, Proposition 7.1

Date. 2026-07-07 **Verdict.** **PASS**

Statement audited

$\max_{p+q=n} \Phi_\tau(S_{p,q}) = \left\lfloor \frac{(2n+1)^2}{24} \right\rfloor$, attained in the branch $q \geq p-1$, and the nonsaturated branch $q \leq p-1$ never exceeds this value.

Method

Exact rational/integer arithmetic (no floating LP) using the branch formulas of Corollary 6.3, for every n with $1 \leq n \leq 3000$; the nonsaturated-branch values are checked against the floor, and the gap $(8n+1)/24 > 1$ is verified for $n \geq 3$.

Verbatim output

```
n range: 1..3000
max_p Phi_tau(S_p,n-p) == floor((2n+1)^2/24) : mismatches = 0
nonsaturated-branch values exceeding floor      : 0
n | floor((2n+1)^2/24) | max_p Phi | argmax p
 3 |          2 |          2.0 | 1
10 |         18 |         18.0 | 4
30 |        155 |        155.0 | 10
99 |       1650 |       1650.0 | 33
300 |      15050 |      15050.0 | 100
1000 |     166833 |     166833.0 | 334
3000 |    1500500 |    1500500.0 | 1000
(8n+1)/24 > 1 for n>=3 (nonsaturated gap): True
RESULT: PASS
```

Reproducibility

The exact script `control.py` and its verbatim output `results.txt` are included in this folder. Re-running `python3 control.py` reproduces `results.txt`. This certificate reports a computational check on finite instances; it is supporting evidence, not a substitute for the analytic proof in Paper II. File integrity is recorded in `SHA256SUMS.txt` (this folder) and in the master `SHA256SUMS.txt` of the package.